IMAGES IN CLINICAL MEDICINE

Spironolactone-Induced Gynecomastia

A 76-YEAR-OLD MAN WITH CORONARY ARTERY DISEASE AND HEART FAILURE with reduced ejection fraction presented to the cardiology clinic with an 8-month history of progressive breast enlargement and tenderness. Spironolactone had been started 4 years before presentation at a dose of 25 mg daily and was increased to 100 mg daily 1 year before presentation. Owing to hyperkalemia, the dose had been reduced to 25 mg approximately 2 months before presentation. Physical examination was notable for symmetric enlargement of glandular breast tissue on both sides. On palpation, there was tenderness but no nodules, nipple discharge, or skin retraction. There was also a sternotomy scar from previous coronary-artery bypass grafting. Laboratory studies showed normal kidney and liver function. A serum testosterone level was low-normal and serum levels of human chorionic gonadotropin, sex hormone–binding globulin, luteinizing hormone, estradiol, and thyrotropin were normal. A diagnosis of spironolactone-induced gynecomastia — an adverse drug effect seen more frequently in men taking more than 100 mg per day — was made. The mechanism is multifactorial and includes androgen-receptor blockade and increased peripheral conversion of testosterone to estradiol. Spironolactone was discontinued and eplerenone — a more selective mineralocorticoid receptor antagonist — was started. At 3 months of follow-up, the patient's breast tenderness had abated but the gynecomastia remained.

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